End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name:<B.Tech>, Semester:<8>
(May, 2025)

Subject Code: BAI 416	**Subject: AI in Healthcare**
Time :3 Hours	**Maximum Marks :60**

Note: Q1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1		(2.5*8=20)	CO Mapping
	a) What are the differences between structured and unstructured healthcare data? Provide an example of each.		**CO1**
	b) Why is data preprocessing essential in healthcare AI applications? Mention two key steps involved.		**CO1**
	c) What are Clinical Decision Support Systems (CDSS), and how do they assist healthcare professionals?		**CO2**
	d) Discuss one ethical concern associated with AI-powered diagnostic tools in healthcare.		**CO2**
	e) What role does Natural Language Processing (NLP) play in Electronic Health Record (EHR) analysis?		**CO1**
	f) Describe one predictive modeling technique used in healthcare for patient risk assessment.		**CO3**
	g) How HIPAA compliance ensures the privacy and security of patient data in AI applications.		**CO4**
	h) Identify one emerging trend in AI-driven healthcare and discuss its potential impact.		**CO4**
UNIT I			**CO Mapping**
Q2	Describe the importance of data preprocessing in AI-driven healthcare applications. Illustrate with an example of data cleaning and transformation.	**(10)**	**CO1**
Q3	Explain the key techniques used for extracting meaningful insights in a step-by-step hands-on analysis of a healthcare dataset.	**(10)**	**CO1**
UNIT II			**CO Mapping**
Q4	Discuss the ethical considerations involved in AI-driven clinical decision support, including biases and patient safety concerns.	**(10)**	**CO2**
Q5	What are the key challenges faced when implementing AI in medical imaging? Discuss real-world limitations and possible solutions	**(10)**	**CO2**
UNIT III			**CO Mapping**
Q6	Discuss various predictive modeling techniques used in healthcare. Compare their advantages and disadvantages.	**(10)**	**CO3**
Q7	Explain the process of validating and evaluating predictive models in healthcare AI applications. Discuss key performance metrics used to evaluate the models.	**(10)**	**CO3**
UNIT IV			**CO Mapping**
Q8	Discuss the ethical guidelines that govern AI applications in healthcare. How can AI-related ethical dilemmas be addressed?	**(10)**	**CO4**
Q9	Analyze the impact of AI on patient data security. What are the major privacy concerns, and how can they be mitigated?	**(10)**	**CO4**